ME571 Final Project – Spring 2023

Reynold's Number Calculation and Experiment Using Solidworks Flow Simulation

Bailey Walke

Introduction

Laminar vs. turbulent flow has been an important concept in this course and is very applicable to real life situations. One way to measure whether flow is laminar or turbulent is using Reynold's Number.

The Reynold's Number is the ratio of the inertial force and shearing force of a liquid. This number is affected by the diameter of the vessel, length of vessel, viscosity and density of the fluid, and the speed of the fluid. As Reynold's number increases, the flow becomes more turbulent. Turbulent flow occurs when inertial forces are dominant, which occurs at faster speeds and with less viscous fluids. Laminar flow correlates to slow moving or viscous fluid when viscous forces are more dominant. This experiment specifically focuses on pipe flow and the analysis of the velocity gradients.

This project utilizes Solidworks Flow Simulation to compare simulated data to calculations made with computational fluid mechanics equations. Different variables are changed to observe the effects of each variable to the Reynold's Number calculation.

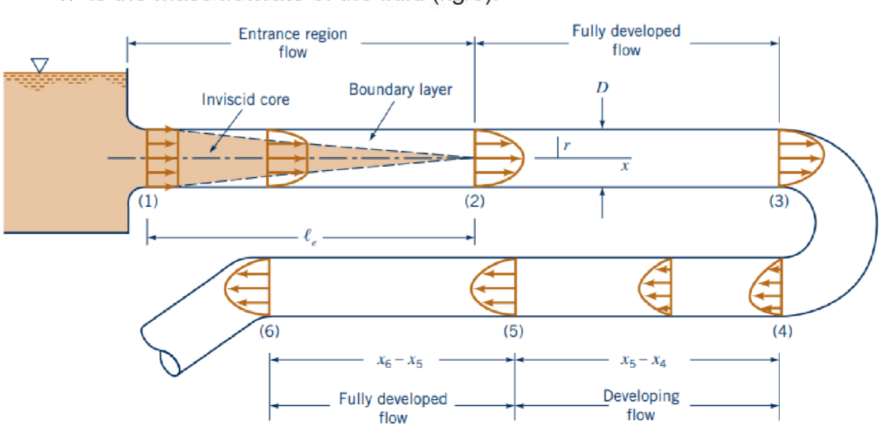
Formulas

For flow in a pipe or tube, the Reynolds number is generally defined as[8]

$$\mathrm{Re} = \frac{uD_\mathrm{H}}{\nu} = \frac{\rho u D_\mathrm{H}}{\mu} = \frac{\rho Q D_\mathrm{H}}{\mu A} = \frac{W D_\mathrm{H}}{\mu A},$$

where

- D_H is the hydraulic diameter of the pipe (the inside diameter if the pipe is circular) (m),
- Q is the volumetric flow rate (m^3/s),
- A is the pipe's *cross-sectional* area ($A = \frac{\pi D^2}{4}$) (m^2),
- u is the mean velocity of the fluid (m/s),
- μ (mu) is the dynamic viscosity of the fluid (Pa·s = N·s/m^2 = kg/(m·s)),
- ν (nu) is the kinematic viscosity ($\nu = \frac{\mu}{\rho}$) (m^2/s),
- ρ (rho) is the density of the fluid (kg/m^3),
- W is the mass flowrate of the fluid (kg/s).

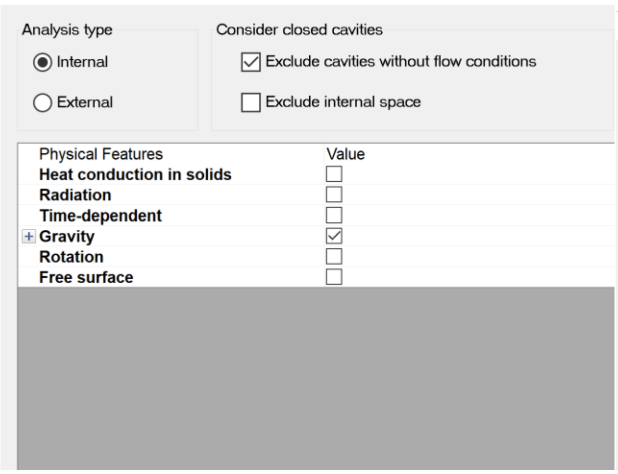


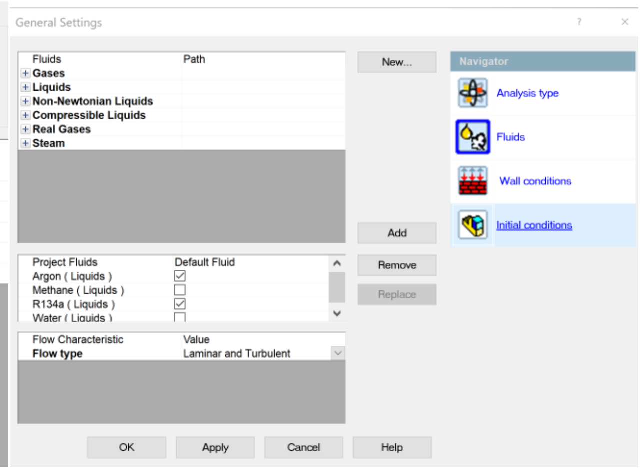
Procedure

A model of a pipe was created in Solidworks with an inner diameter of 200 mm and a length of 5 meters.

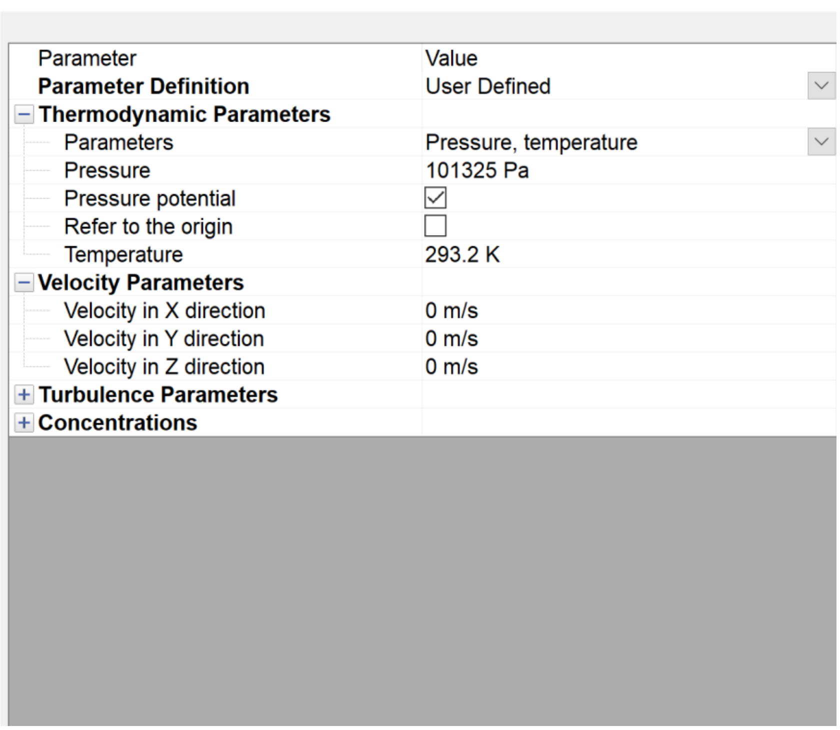
Solidworks Flow Simulation is used to create simulations. 6 different simulations are created with one variable changes for each iteration. Boundary conditions remain the same for each iteration. The tube walls are adiabatic and smooth. The initial volumetric flowrate is set at 1 m^3/s for the inlet and a pressure outlet.

General Settings



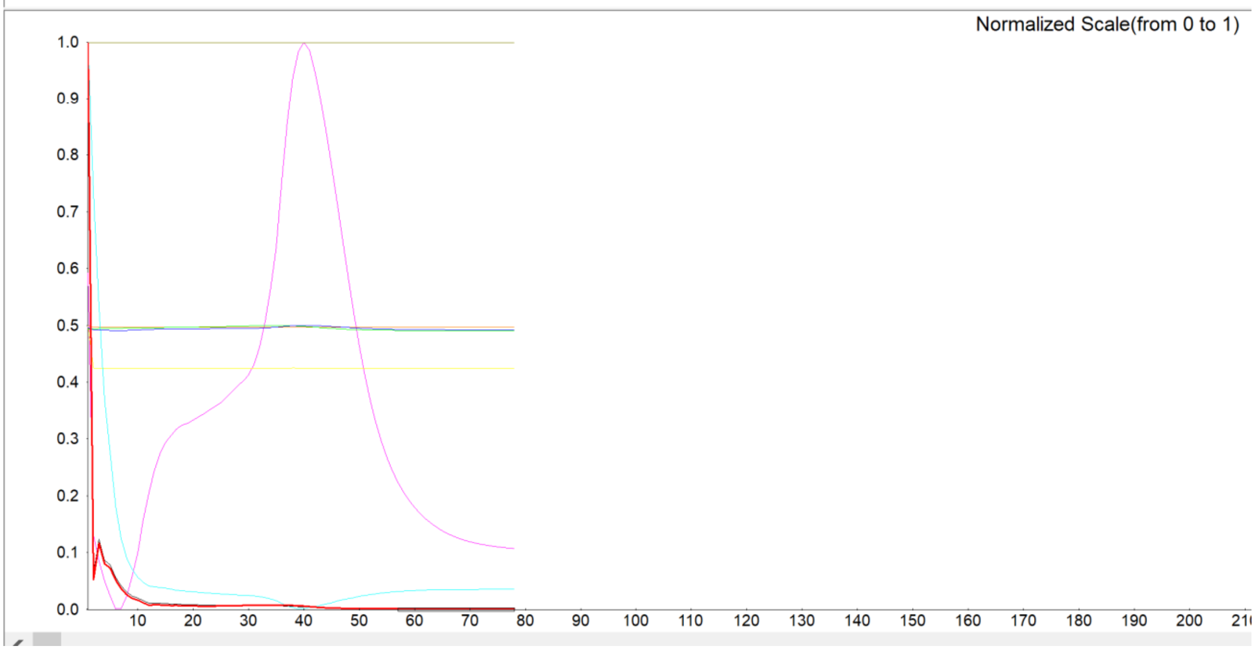


General Settings



Convergence of data:

Name	Current Value	Progress	Criterion	Averaged Value
GG Average Total Pressure 1	772319 Pa	Achieved (IT = 41)	194345 Pa	772433 Pa
GG Mass (Fluid) 2	183.976 kg	Achieved (IT = 41)	1.83964 k	183.976 kg
GG Mass Flow Rate 3	3.08873e-05 kg/s	Achieved (IT = 41)	23.8883 k	-3.75171e-05 kg/s
GG Volume Flow Rate 4	7.91817e-06 m^3/s	Achieved (IT = 41)	0.0200012	1.31453e-05 m^3/s
SG Environment Pressure 2 Mass Flow Rate	-1194.42 kg/s	Achieved (IT = 41)	1.19442 k	-1194.42 kg/s
SG Environment Pressure 2 Velocity Av	32.1355 m/s	Achieved (IT = 41)	0.0086650	32.1354 m/s
SG Environment Pressure 2 Volume Flow Rate	-0.999992 m^3/s	Achieved (IT = 78)	1.93384e-	-0.999987 m^3/s
SG Inlet Volume Flow 1 Static Pressure Av	212598 Pa	Achieved (IT = 41)	419872 Pa	212621 Pa
SG Inlet Volume Flow 1 Total Pressure Av	829341 Pa	Achieved (IT = 41)	419872 Pa	829365 Pa
SG Inlet Volume Flow 1 Velocity Av	32.1358 m/s	Achieved (IT = 41)	3.21358e-	32.1358 m/s



Data

<u>Part 1: Changing Flowrate</u>

With the three iterations of volumetric flowrates of 1, 5 and 10 m^3/s, the velocity plots were below where generated.

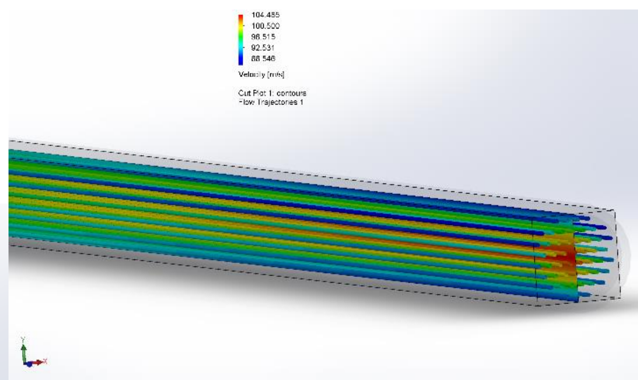
At 18 degrees Celsius, water has a kinematic viscosity of $1.0533*10^{-6}$ m^2/s, dynamic viscosity of $1.0518 *10^{-3}$ Ns/m2, and its density is 1,000 kg/m^3. The maximum average velocity of each iteration is what is used in Reynold’s number calculation. The diameter of the pipe is 0.2 meters.

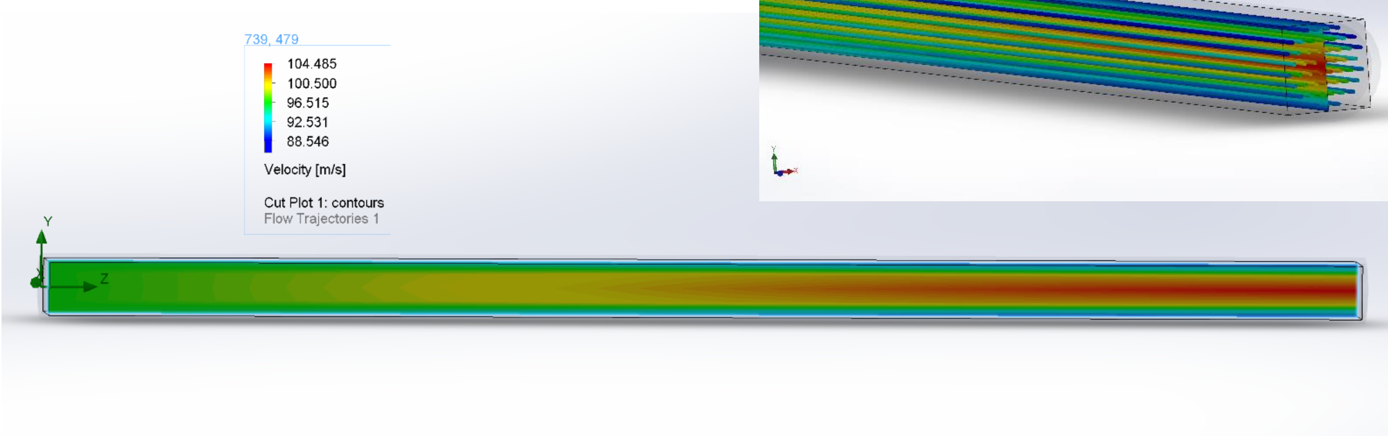
Q = 1 m^3/s

Theoretical: $\mathrm{Re} = \frac{\rho Q D_\mathrm{H}}{\mu A}$ = (1000)(1)(0.2)/(1.0518*10^{-3})(pi/4*(.2)^2) = 6052669

Experimental: $u_{av\ max}$ = 104.455 m/s

$$\mathrm{Re} = \frac{u D_\mathrm{H}}{\nu}$$ = (104.455)(.2)/(1.0533*10^{-6}) = 19833856



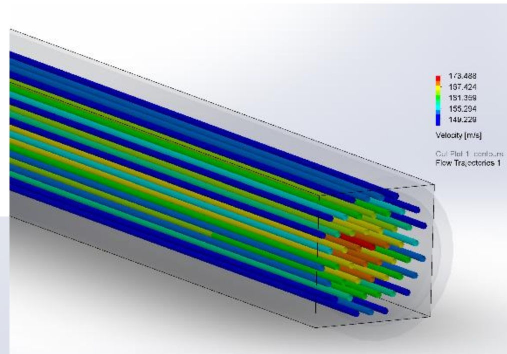


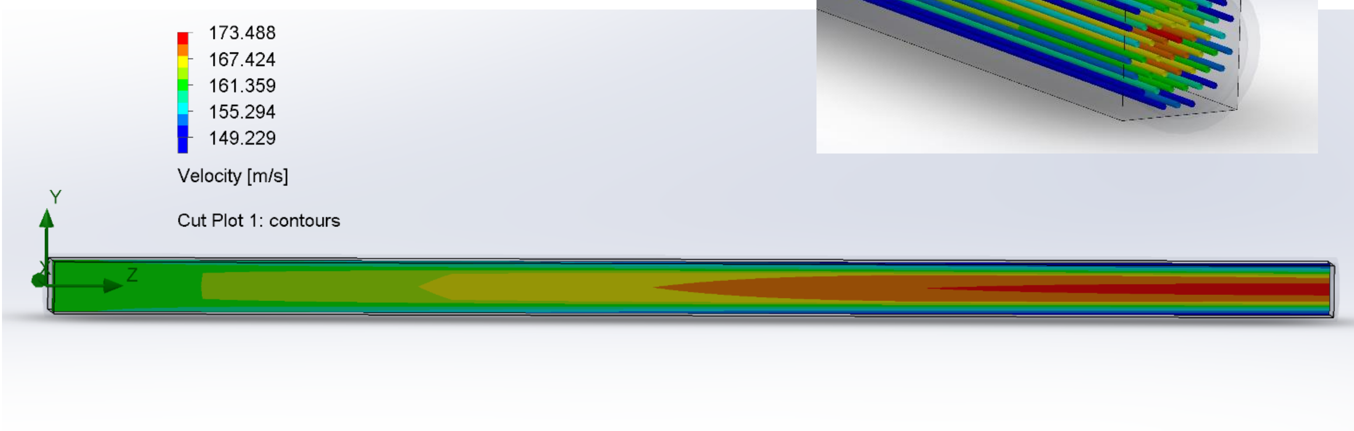
Q = 5 m^3/s

Theoretical: Re $= \frac{\rho Q D_{\mathrm{H}}}{\mu A}$ = (1000)(5)(0.2)/(1.0518*10^{-3})(pi/4*(.2)^2) = 30263347

Experimental: $u_{av\ max}$ = 173.488 m/s

$$\mathrm{Re} = \frac{u D_{\mathrm{H}}}{\nu}$$ = (173.488)(.2)/(1.0533*10^{-6}) = 32941801



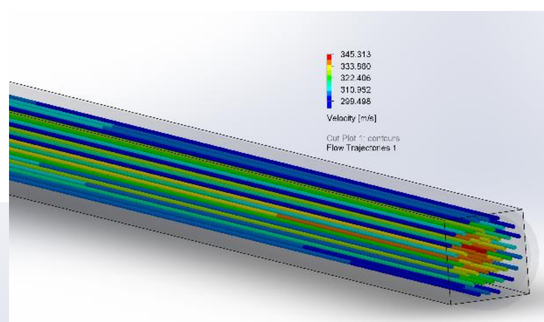


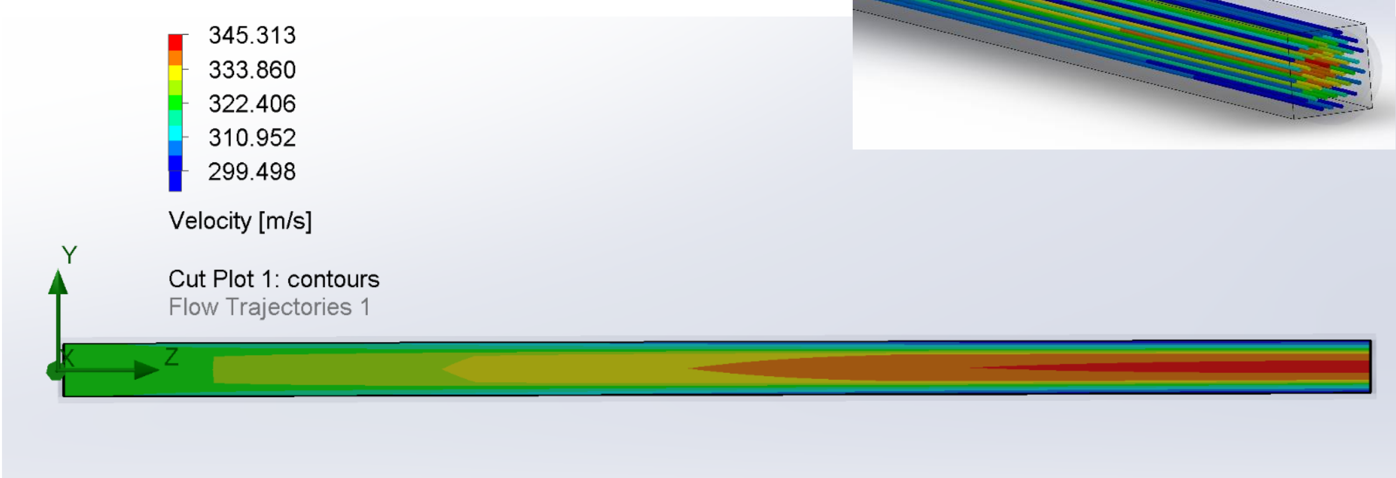
Q = 10 m^3/s

Theoretical: Re $= \frac{\rho Q D_{\mathrm{H}}}{\mu A}$ = (1000)(10)(0.2)/(1.0518*10^{-3})(pi/4*(.2)^2) = 60526694.5

Experimental: $u_{av\ max}$ = 345.313 m/s

$$\mathrm{Re} = \frac{u D_{\mathrm{H}}}{\nu}$$ = (345.313)(.2)/(1.0533*10^{-6}) = 65567834





Part 2: Changing Viscosity

With the three iterations of different fluids methane, argon, and R134a, the velocity plots were below where generated.

The maximum average velocity of each iteration is what is used in the Reynold's number calculation. The diameter of the pipe is 0.2 meters.

Methane (Q = 1m^3/s)

Dynamic viscosity = 10.84 * 10^{-6} Pa*s

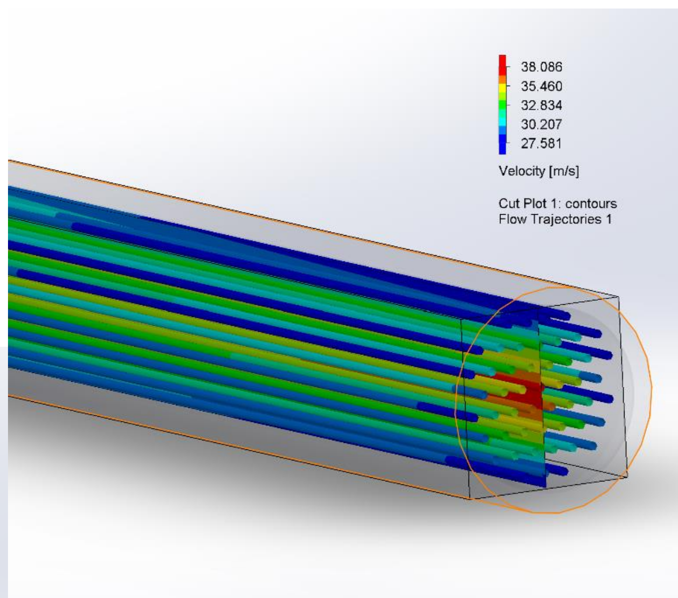
Kinetic viscosity = 16.33 * 10^{-6} m^2/s

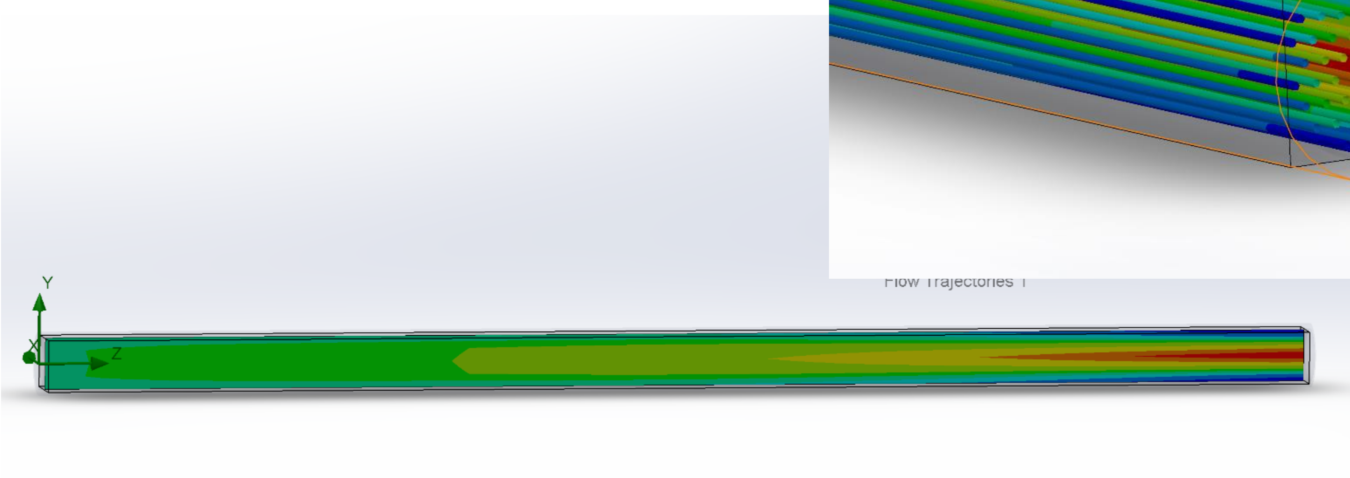
Density = 0.657 kg/m^3

Theotretical: R $= \frac{\rho Q D_{\mathrm{H}}}{\mu A}$ = (0.657)(1)(0.2)/(10.84 * 10^{-6})(pi/4*(.2)^2) = 385848

Experimental: $u_{av\ max}$ = 38.086 m/s

$\mathrm{Re} = \frac{u D_{\mathrm{H}}}{\nu}$ = (38.086)(0.2)/(16.33 * 10^{-6}) = 466454





Argon (liquid) (Q = 1m^3/s)

Dynamic viscosity = 270.7 * 10^{-6} Pa*s

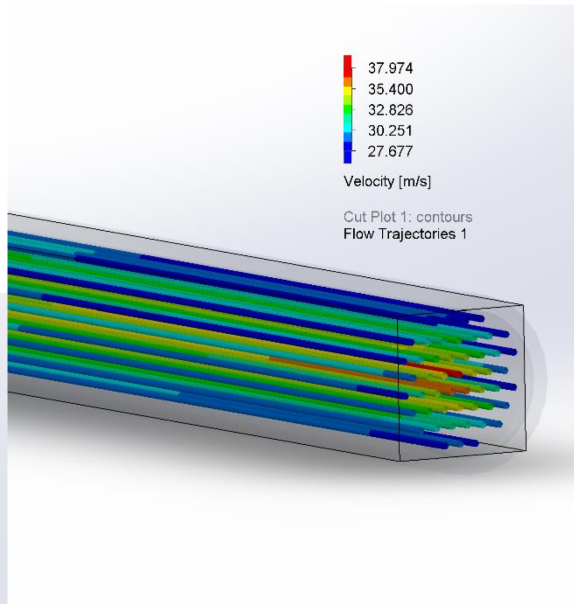
Kinetic viscosity = 194.75 * 10^{-6} m^2/s

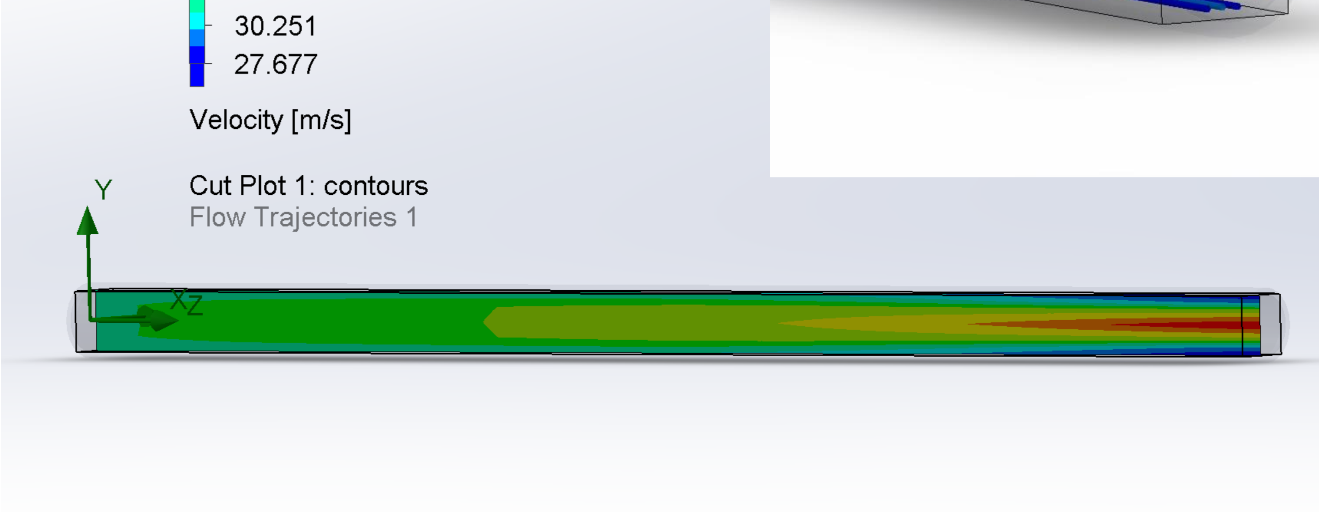
Density = 1.39 kg/m^3

Theotretical: R $= \frac{\rho Q D_{\mathrm{H}}}{\mu A}$ = (1.39)(1)(0.2)/(270.7 * 10^{-6})(pi/4*(.2)^2) = 32689

Experimental: $u_{av\ max}$ = 37.974 m/s

$$\mathrm{Re} = \frac{u D_{\mathrm{H}}}{\nu}$$ = (37.974)(0.2)/(194.75 * 10^{-6}) = 38998





R134a (liquid) (Q = 1m^3/s)

Dynamic viscosity = .4298 * 10^3 Pa*s

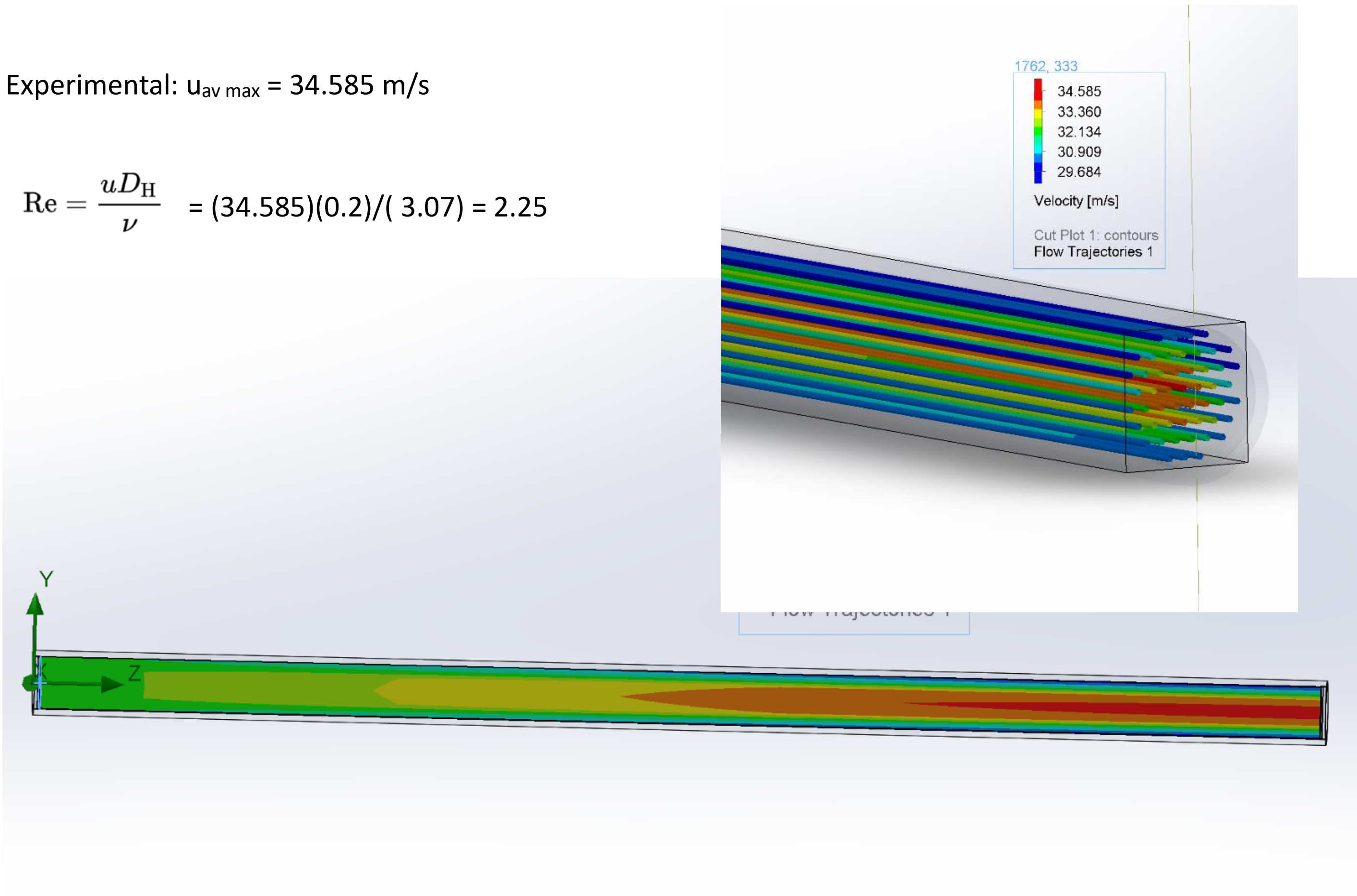
Kinetic viscosity = 3.07 m^2/s

Density = 1400 kg/m^3

Theotretical: Re $= \frac{\rho Q D_{\mathrm{H}}}{\mu A}$ = (1400)(1)(0.2)/(.4298 * 10^3)(pi/4*(.2)^2) = 13.5

Experimental: $u_{av\ max}$ = 34.585 m/s

$\mathrm{Re} = \frac{u D_{\mathrm{H}}}{\nu}$ = (34.585)(0.2)/(3.07) = 2.25



Analysis/Discussion

Most of the flows calculated appear to be turbulent. I did not expect this when conducting the experiment because the flow rates seemed to be fairly low, but considering the size of the pipe, it makes sense that the flow is turbulent.

The calculated error between the theoretical and experimental Reynold's Number calculation mostly stayed above 80%, which means the simulation was fairly accurate at modeling velocity.

When looking at the cut plots, I noticed interesting patterns with the red/orange areas. For the first iteration of the flowrate experiment, the cut plot had a much less "harsh" transition from green to red, compared to the 5m^3/s and 10m^3/s. Higher velocities were also reached sooner from the higher flowrates. R134a has a higher viscosity compared to water, argon, and methane, and it met its higher velocities sooner than the other fluids. I feel like this relates to fully developed flow or developing flow, which relates to speed and viscosity. I could see a similar shape to the velocity field as what was shown in the cut plots.

Conclusion

In conclusion, this project helped me learn how to use Solidworks flow simulation and how to create my own experiment regarding fluid mechanics. I got to have a more hands-on experience with laminar/turbulent flow by changing variables and visually seeing how the velocity plots were affected.

References

Reynolds number - Wikipedia

Water - Dynamic (Absolute) and Kinematic Viscosity vs. Temperature and Pressure (engineeringtoolbox.com)

The Viscosity and Thermal Conductivity Coefficients of Gaseous and Liquid Argon | Journal of Physical and Chemical Reference Data | AIP Publishing

(PDF) The viscosity of liquid R134a (researchgate.net)

Water - Dynamic (Absolute) and Kinematic Viscosity vs. Temperature and Pressure (engineeringtoolbox.com)

Learn Step by Step How to do Flow Simulation in SolidWorks on Cross Flow Turbine - YouTube

Solidworks flow simulation basic: Laminar pipe flow - YouTube

Internal Pipe Flow with SOLIDWORKS Flow Simulation - YouTube